Cable-Fault Tolerance in Decentralized Cooperative Aerial Transport: N-Agnostic Architecture, Evidence, and Open Bounds

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***Abstract*—When a cable snaps during multi-quadrotor cooperative payload transport, the payload drops, the freed agent accelerates away, and every surviving agent faces an instantaneous load-share jump—a 50% increase for $N{=}3$. This paper shows that a decentralized, communication-free architecture absorbs such faults with no detection, no reconfiguration, and no inter-agent communication.**

The mechanism is a structural property of N-agnostic GPAC (Geometric Position and Attitude Control): because each agent's PID-plus-gravity-feedforward law uses only local measurements with fixed gains, the per-agent error dynamics share the same system matrices regardless of cable topology (Proposition 1). Cable snaps change the disturbance input but not the system. This topology invariance yields two closed-form design tools: a fault-tolerance cost ratio $\rho_{\mathrm{FT}} = N/(N{-}k)$ for actuator sizing and a finite-time transient bound whose scaling is confirmed to be topology-independent across scenarios. Ablation identifies PID feedback with gravity feedforward—not cable-tension feedforward ($<1\%$ RMSE contribution)—as the primary fault-absorption mechanism.

Full-Drake multibody simulations across $N \in \{3,5,7\}$ with up to three cascading cable snaps demonstrate payload RMSE of 43.6–48.8 cm, *decreasing* with team size despite absorbing proportionally more faults. A centralized oracle with perfect fault knowledge improves over the passive architecture by only 0.7%, showing that the price of decentralization under cable faults is effectively zero. Results are validated under sensor-realistic conditions (ESKF + Dryden wind), Monte Carlo campaigns (30 runs, $<3\%$ RMSE variation), and a thrust-saturation boundary sweep identifying a sharp phase transition at the theoretical minimum with $3\times$ headroom.

The topology-invariant error structure is proven for a simplified per-agent model. Extension to the full nonlinear bead-chain plant is supported by simulation evidence and the ISS switched-system framework; the formal stability proof remains an open problem.

***Index Terms*—Cooperative aerial transport, cable-fault tolerance, N-agnostic control, PID fault tolerance, topology invariance, multi-quadrotor systems**

I. Introduction

In multi-quadrotor cooperative cable-suspended payload transport, N underactuated vehicles are tethered to a shared rigid payload through flexible cables, forming a system on the configuration manifold $\mathcal{Q} = \mathrm{SE}(3) \times (\mathrm{SE}(3) \times \mathbb{S}^2)^N$ with $6 + 8N$ degrees of freedom. Geometric control methods [1]–[4] provide almost-global exponential stability on $\mathrm{SO}(3)$ and enable aggressive trajectory tracking with cable-suspended loads, bringing multi-vehicle cable transport close to deployment readiness [5], [6]. Yet every cable is a single point of mechanical failure. When a cable snaps, three coupled events occur simultaneously: the payload accelerates downward ($\approx$3.3 m/s^2), the freed agent accelerates upward ($\approx$6.5 m/s^2), and every remaining agent experiences an instantaneous load-share jump—a 50% increase for $N{=}3$. These effects propagate on timescales from sub-10 ms (rigid-body force transmission) to $\sim$200 ms (cable wave propagation), creating a multi-channel fault signature that each surviving agent must absorb without centralized coordination or inter-agent communication.

The difficulty is compounded by the architecture's defining property. The baseline GPAC (Geometric Position and Attitude Control) hierarchy enforces an *N-agnostic constraint*: each agent knows only its own mass, inertia, local sensor measurements, cable tension and direction, and a shared reference trajectory. It knows neither the team size, the payload mass, nor any other agent's state. This constraint eliminates single points of failure and communication dependency, making the architecture scalable—but it also blocks model-based fault detection methods such as multiple-model adaptive estimation (which requires $N{+}1$ hypothesis filters). The tension-channel signal-to-noise ratio for a remote cable break is approximately 1.3, indistinguishable from wind perturbations. Cable-fault tolerance in this setting must therefore work without detecting, isolating, or recovering from the fault.

Despite mature geometric transport theory and extensive UAV fault tolerance work for actuator and sensor failures [7], [8], no published work addresses cable-fault tolerance in cooperative cable-suspended aerial transport. The only adjacent result, Liang et al. [9], assumes known failure identity and centralized reconfiguration—both incompatible with N-agnostic architectures. Zeng et al. [10] and Gao et al. [11] observed implicit resilience to agent loss in learning-based transport but did not formalize the mechanism, and neither addressed cable-suspended underactuated systems.

The research objective was shaped by an empirical observation. The predecessor system Tether_Lift [12] established that the GPAC hierarchy can coordinate cooperative lift using only local measurements under the N-agnostic constraint. The present paper (Tether_Grace) asks whether that same architectural property also yields resilience to abrupt cable-topology change.

The answer lies in PID feedback with gravity feedforward. The existing GPAC baseline, evaluated under cable-snap faults with no explicit fault handling, exhibited stable post-fault tracking with payload RMSE of 44–49 cm across $N \in \{3,5,7\}$, decreasing with team size despite cascading faults. The mechanism is the N-agnostic PID structure: each agent computes gravity feedforward from its own mass, PID feedback from its own position error, and anti-swing damping from local cable angular velocity. None of these terms depends on N, m_L, or any other agent's state. When a cable snaps,

the surviving agents' tracking errors update instantaneously through the physics, and the PID integral term absorbs the increased load share without adaptation or re-convergence delay. Ablation confirms that disabling cable-tension feedforward changes RMSE by less than 1% (Section VI-C).

This transforms the research problem from *how to detect and recover from cable faults* to *whether the emergent fault tolerance can be formalized and quantified*: under what conditions does an N-agnostic controller with local cable-tension feedforward produce bounded tracking degradation across all cable topologies? Figure 1 summarizes this narrowing from the broader Tether_Lift baseline to the present Tether_Grace fault-tolerance question and its current evidence boundary.

This paper addresses that question by identifying the structural property that makes the architecture topology-invariant, deriving operationally useful bounds, and validating them with comprehensive simulation evidence. The replayed full-Drake path uses the GPAC hierarchy with PID-plus-feedforward control. A post-fault geometry redistribution path is additionally described. Adaptive extensions ($\mathcal{L}_1$, concurrent learning) are implemented in the codebase but remain future integration work.

Two structural properties make the PID-plus-gravity-feedforward design attractive. First, PID feedback with gravity feedforward provides the core fault-absorption capability (confirmed by ablation, Section VI-C). Second, all controller terms are functions of the agent's own state, so no agent's control law can be invalidated by a topology change it cannot observe.

These properties motivate a *topology-invariance design hypothesis*: if the post-fault disturbance remains inside a fault-inclusive bound and the local thrust margin is adequate, then the tracking bound is governed by controller bandwidth and disturbance size rather than cable identity. Full-Drake multibody results provide strong numerical support: under matched single-fault conditions (identical fault, varying N), payload RMSE decreases from 48.8 cm to 38.4 cm for $N{=}3$ to $N{=}7$. A thrust-saturation boundary sweep further identifies a sharp phase transition at $f_{\max} \approx 45$ N that coincides with the theoretical post-fault hover minimum, with the default 150 N providing $3\times$ headroom.

A. Contributions

C1 *Topology-invariant error structure and mechanism identification:* A structural analysis (Proposition 1) showing that the N-agnostic PID-plus-feedforward control law produces per-agent error dynamics that are identical across all cable topologies—cable snaps change the disturbance input but not the system matrices. This structural property motivates a topology-invariance hypothesis that is validated by full-Drake simulation. A finite-time transient prediction ($\sim$18 cm predicted, observed $\sim$107 cm; the $\sim 5.8\times$ ratio is topology-invariant across scenarios) and a closed-form cost ratio $\rho_{\mathrm{FT}} = N/(N{-}k)$ provide design criteria for actuator sizing. Ablation confirms PID feedback with gravity feedforward as the primary mechanism (cable-tension feedforward $< 1\%$ RMSE change).

C2 *Comprehensive multibody evidence:* Full-Drake simulation across $N \in \{3, 5, 7\}$ with 1–3 cascading cable-snap faults under both ideal and sensor-realistic (ESKF + Dryden wind) conditions. Monte Carlo validation (30 runs per scenario) with randomized fault timing, wind, cable lengths, and initial conditions. Matched single-fault experiments ($48.8 \rightarrow 38.4$ cm), fault-count isolation, trajectory generalization, and baseline comparison provide comprehensive deconfounding.

C3 *Quantitative design framework:* A fault-tolerance cost ratio ρ_{FT} validated against empirical nominal-vs-fault RMSE ratio, a thrust-margin screening methodology validated by the $f_{\max}$ phase transition at the theoretical minimum, and a tilt-corrected capacity analysis for actuator sizing.

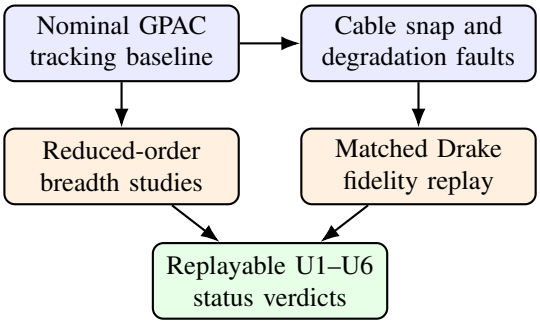


Fig. 1: Overview of the paper scope. The study begins from the validated GPAC baseline (Tether_Lift), introduces the cable-snap fault problem, documents the Tether_Grace fault-tolerance extension, and evaluates the present evidence.

B. Organization

Section II reviews related work. Section III defines the system model, the cable-snap fault event, and the formal problem. Section IV presents the fault-tolerant architecture and its integration boundaries. Section V defines the evidence methodology. Section VI reports simulation results. Section VII interprets findings, identifies gaps, and concludes.

II. Related Works

This section reviews three research areas whose intersection defines the paper's contribution: cooperative cable-suspended aerial transport (Section II-A), fault-tolerant control for UAVs (Section II-B), and adaptive control relevant to topology-change resilience (Section II-C). Section II-D reviews safety-critical control tools, Section II-E reviews emergent resilience in multi-agent systems, and Section II-F synthesizes the gap.

A. Cooperative Cable-Suspended Aerial Transport

The geometric control framework originates with Lee, Leok, and McClamroch [1], who developed singularity-free tracking on $\mathrm{SO}(3)$. Sreenath, Lee, and Kumar [2] extended this to cooperative cable-suspended payload manipulation on $\mathrm{SE}(3) \times (\mathbb{S}^2)^N$. Sreenath and Kumar [13] established differential flatness for single-quadrotor cable-suspended loads. Lee [3] proved almost-global exponential stability on $\mathrm{SE}(3) \times \mathbb{S}^2$ for cable-suspended rigid body transport. Dai, Lee, and

Bernstein [4] introduced adaptive load mass estimation, and Goodarzi, Lee, and Lee [14] extended geometric stabilization to flexible cables.

Recent advances include Sun et al. [6], who demonstrated agile cooperative aerial manipulation with INDI-based control and centralized NMPC, achieving $8\times$ greater acceleration than prior methods (Science Robotics, 2025). Wahba and Hönig [15] developed efficient QP-based cable force allocation for embedded microcontrollers. Li, Liu, and Loianno [16] created RotorTM, a validated flexible simulator. Costantini, de Angelis, and Giulietti [17] used two-time-scale singular perturbation for trajectory-swing separation. Additional contributions include cooperative transport via monocular vision [18], hierarchical control via the attractive ellipsoid method [19], adaptive RISE control [20], and a comprehensive field survey [5].

Every published controller assumes all cables remain intact. No result addresses tracking guarantees under cable loss, no design criterion exists for sizing margins against cable-loss disturbances, and no simulation study characterizes fault-tolerance behavior in this system class.

B. Fault-Tolerant Control and Fault Detection for UAVs

The UAV fault-tolerant control (FTC) literature is extensive for actuator and sensor faults. Sun et al. [21] demonstrated INDI-based flight surviving complete loss of two opposing rotors. Zhang et al. [22] proposed fault-tolerant neural CBFs for sensor faults. Chen et al. [23] trained a single RL policy handling arbitrary single-rotor failures without detection. Adaika et al. [7] surveyed 32 publications on fault detection and diagnosis for UAVs. Yu et al. [8] reviewed fault-tolerant cooperative control of multiple UAVs.

Critical observation: All studied fault modes are actuator faults (rotor degradation/loss) or sensor faults (IMU bias, GPS dropout). Cable faults—which change the mechanical coupling between agents and payload—are entirely absent.

The only paper addressing cable-related failure in multi-UAV transport is Liang et al. [9], who propose dynamic load reassignment with sliding-mode observers. However, this work assumes known failure identity, requires centralized reconfiguration, provides no detection mechanism, and includes no experimental validation—all incompatible with the N-agnostic setting.

In cable-driven parallel robots, Raman et al. [24] addressed cable failure using multiple-model estimation, Boschetti, Minto, and Trevisani [25] proposed sensorless failure detection via digital twins, and Zhu et al. [26] developed interval observers for a single quadrotor with a cable-suspended load. None addresses cooperative multi-UAV transport or satisfies the N-agnostic constraint.

C. Adaptive Control and Disturbance Estimation

Two adaptive paradigms are directly relevant to topology-change resilience. Concurrent learning (CL) [27]–[29] replays stored regressor-measurement pairs to achieve parameter convergence without persistent excitation, but its data buffer becomes a liability after abrupt parameter changes as pre-change data biases the estimate during buffer turnover. $\mathcal{L}_1$ adaptive control [30], [31] provides uniform transient guarantees with memoryless piecewise-constant adaptation, making it structurally suited to topology-change scenarios; Wu et al. [32] demonstrated $\mathcal{L}_1$ augmentation of geometric quadrotor control at $>$200 Hz. In decentralized adaptive transport, Barawkar et al. [33] used fuzzy logic, Bai and Wen [34] developed force-consensus control, Gao et al. [11] demonstrated adaptation to robot departure, and Wang et al. [35] introduced Auto-Multilift.

Neither CL nor $\mathcal{L}_1$ has been applied to cooperative cable transport, and neither theory has been analyzed under abrupt topology changes where the system structure—not just parameter values—changes instantaneously.

D. Safety-Critical Control

Control barrier functions (CBFs) [36], [37] provide forward-invariance via QP-based control modification, with Input-to-State Safety (ISSf) [38] extending CBF guarantees under bounded disturbances and high-order extensions [39] handling relative-degree constraints. Yang et al. [40] combined CBFs with disturbance estimators for cable-suspended payload safety. Complementary tracking-error bounds from contraction theory [41], [42] and tube-based motion planning [43] provide relevant tools. Neither CBF nor contraction-based methods have been applied to post-fault topology-change scenarios in multi-vehicle cable transport.

E. Emergent Resilience and Switched Systems

The theoretical foundations for analyzing topology changes include stability of switched systems under average dwell-time [44], ISS under switching [45], mode-dependent average dwell time [46], and the hybrid dynamical systems framework [47] for continuous flow interrupted by discrete jumps—the natural setting for cable-snap events.

Two empirical observations of emergent fault tolerance are closest to this work. Zeng et al. [10] demonstrated implicit resilience to MAV loss in multi-agent RL for cable-suspended transport with zero-shot sim-to-real transfer, but did not formalize the mechanism. Gao et al. [11] showed adaptation to robot departure, but used fully actuated hexarotors with commanded departure rather than spontaneous cable faults.

No work formalizes the mechanism by which N-agnostic control provides implicit cable-fault tolerance. The switched systems literature provides relevant tools but has not been applied to N-agnostic cable-suspended transport with instantaneous, undetected topology changes.

F. Summary of the Identified Gap

The contribution lies at the intersection of three well-populated but previously unconnected areas:

(i) *Cooperative cable-suspended transport* [1]–[6], [15], [16]: mature geometric control with no cable-fault handling.

(ii) *UAV fault-tolerant control* [7], [8], [21]–[23]: extensive actuator/sensor fault work, but cable faults are absent.

(iii) *Adaptive decentralized control* [27], [30]–[33]: rich theory not analyzed for topology-change resilience in cable transport.

No prior work provides tracking bounds or design criteria for cable-fault tolerance in decentralized cooperative aerial transport. The present paper addresses this gap.

III. Problem Statement

This section defines the physical system, the baseline control architecture, the cable-snap fault event, and the formal problem.

A. System Model

1) Configuration Space: The system comprises N quadrotors cooperatively transporting a rigid payload via flexible cables. The state evolves on

$$\mathcal{Q} = \mathrm{SE}(3) \times \left(\mathrm{SE}(3) \times \mathbb{S}^2\right)^N, \quad \dim(\mathcal{Q}) = 6 + 8N, \quad (1)$$

where $\mathrm{SE}(3)$ captures the payload's six-DOF pose, each quadrotor contributes a further $\mathrm{SE}(3)$, and each cable direction lives on $\mathbb{S}^2$. The default configuration uses $N = 3$ ($\dim(\mathcal{Q}) = 30$), with scaling experiments extending to $N \in \{5, 7\}$.

2) Quadrotor Dynamics: Each quadrotor $i \in \{1, \ldots, N\}$ has mass $m_Q = 1.5$ kg and obeys (with $\boldsymbol{e}_3 = [0, 0, 1]^\top$ pointing upward and $g = 9.81$ m/s^2):

$$m_Q \ddot{\boldsymbol{p}}_i = -m_Q g\, \boldsymbol{e}_3 + f_i R_i \boldsymbol{e}_3 + \boldsymbol{F}_i^{\mathrm{cable}} + \boldsymbol{F}_i^{\mathrm{wind}}, \quad (2)$$

$$J\dot{\boldsymbol{\Omega}}_i = -\boldsymbol{\Omega}_i \times (J\boldsymbol{\Omega}_i) + \boldsymbol{\tau}_i + \boldsymbol{\tau}_i^{\mathrm{ext}}, \quad (3)$$

$$\dot{R}_i = R_i \widehat{\boldsymbol{\Omega}_i}, \quad (4)$$

where $f_i \in [0, f_{\max}]$ is the scalar thrust ($f_{\max} = 150$ N), $R_i \in \mathrm{SO}(3)$ is the body-to-world rotation, $J \in \mathbb{R}^{3\times 3}$ is the inertia tensor (from a $0.30 \times 0.30 \times 0.10$ m solid box), $\boldsymbol{\Omega}_i \in \mathbb{R}^3$ is the body angular velocity, $\boldsymbol{\tau}_i \in \mathbb{R}^3$ is the commanded torque ($|\tau_k| \le 10$ N·m per axis), $\boldsymbol{F}_i^{\mathrm{cable}}$ is the cable force, and $\boldsymbol{F}_i^{\mathrm{wind}}$ is the aerodynamic disturbance.

3) Payload Dynamics: The payload is a rigid sphere of mass $m_L = 3.0$ kg and radius $r_L = 0.15$ m:

$$m_L \ddot{\boldsymbol{p}}_L = -m_L g\, \boldsymbol{e}_3 + \textstyle\sum_{i=1}^N T_i \boldsymbol{q}_i + \boldsymbol{F}^{\mathrm{contact}} + \boldsymbol{F}_L^{\mathrm{wind}}, \quad (5)$$

where $T_i \ge 0$ is the scalar cable tension, $\boldsymbol{q}_i \in \mathbb{S}^2$ is the unit direction from payload to quadrotor i, and $\boldsymbol{F}^{\mathrm{contact}}$ accounts for ground interaction (Coulomb friction, $\mu_s = 0.5$, $\mu_k = 0.3$). Cable attachment points are at radius 0.105 m on the payload surface. The payload mass m_L is *unknown to every agent*. The force balance in (5) is the plant-level relation that later explains how cable-load redistribution appears immediately at the payload during a snap event.

4) Bead-Chain Cable Model: Each cable is modeled as a chain of $n_b = 8$ point-mass beads ($m_{\mathrm{rope}} = 0.2$ kg) creating $n_b + 1 = 9$ spring-damper segments with tension

$$\mathcal{T}_j = \begin{cases} 0, & \text{if } \epsilon_j \le 0 \text{ (slack)}, \\ k_s \epsilon_j + c_s \max(0, \dot{\epsilon}_j), & \text{if } \epsilon_j > 0 \text{ (taut)}, \end{cases} \quad (6)$$

where $\epsilon_j = \|\boldsymbol{b}_{j-1} - \boldsymbol{b}_j\| - L_0$ is the segment stretch and $L_0 = L_{\mathrm{rest}}/(n_b + 1)$. Stiffness is derived from maximum allowable strain $\varepsilon_{\max} = 0.15$:

$$k_s = \frac{m_L g}{N \cdot L_{\mathrm{rest}} \cdot \varepsilon_{\max}} \cdot (n_b + 1). \quad (7)$$

The piecewise law in (6) is the key cable nonlinearity for the reported fault transients, because it captures the slack-to-taut switching that a rigid-cable model would miss. Damping is $c_s = c_{\mathrm{ref}}\sqrt{k_s/k_{\mathrm{ref}}}$ with $c_{\mathrm{ref}} = 15$ N·s/m, $k_{\mathrm{ref}} = 300$ N/m. Cable lengths are sampled from per-quadrotor Gaussians: $L_i \sim \mathcal{N}(\mu_i, s_i^2)$ with means $\mu \in \{1.0, 1.1, 0.95\}$ m and standard deviations $s \in \{0.05, 0.08, 0.06\}$ m, producing up to 19% asymmetry.

The tension-only constraint means cables cannot push. The bead-chain discretization captures lateral vibration, wave propagation ($v_{\mathrm{wave}} \approx 7$ m/s), distributed inertia, and slack-to-taut transitions that rigid-cable assumptions miss. The plant is simulated in Drake [48] at $\Delta t_{\mathrm{sim}} = 0.2$ ms (5 kHz).

5) Wind Disturbance: Aerodynamic disturbance follows the MIL-F-8785C Dryden turbulence model ($\sigma_u = \sigma_v = 1.0$ m/s, $\sigma_w = 0.5$ m/s; $L_u = L_v = 200$ m, $L_w = 50$ m; mean wind $[2.0, 0, 0]$ m/s). Spatial correlation decays as $\rho(\boldsymbol{p}_i, \boldsymbol{p}_j) = \exp(-\|\boldsymbol{p}_i - \boldsymbol{p}_j\|/l_c)$ with $l_c = 10$ m.

6) Sensor Suite and State Estimation: Each agent carries an IMU (400 Hz; gyroscope noise 5×10^{-4} rad/s/$\sqrt{\mathrm{Hz}}$; accelerometer noise 4×10^{-3} m/s^2/$\sqrt{\mathrm{Hz}}$), GPS (10 Hz; position noise $[0.02, 0.02, 0.05]$ m), barometer (25 Hz; 0.3 m white noise, 0.2 m correlated with 5 s time constant), and a cable tension/direction sensor (200 Hz). State estimation uses a 15-state error-state Kalman filter (ESKF) with multiplicative quaternion error, fusing IMU propagation at 400 Hz with discrete GPS and barometer updates.

B. The N-Agnostic Constraint

Each agent i knows only:

- its own mass m_Q and inertia J;
- its ESKF state estimate $(\hat{\boldsymbol{p}}_i, \hat{\boldsymbol{v}}_i, \hat{R}_i)$ and bias estimates;
- its cable tension T_i and direction $\boldsymbol{q}_i$;
- a shared reference trajectory $\boldsymbol{p}_d(t)$ (pre-broadcast, evaluated locally).

It does *not* know: the number of other agents, the payload mass, any other agent's state, or whether other agents are connected. This eliminates single points of failure and communication dependency, making the per-agent policy identical regardless of team size.

C. Baseline GPAC Architecture

The baseline GPAC architecture is a four-layer hierarchical controller per agent with separated timescales.

1) Layer 1: Position and Anti-Swing Control (50 Hz): Computes the desired force:

$$\boldsymbol{F}_{\mathrm{des},i} = \boldsymbol{F}_{\mathrm{fb}} + \boldsymbol{F}_{\mathrm{ff}} + \boldsymbol{F}_{\mathrm{cable}} + \boldsymbol{F}_{\mathrm{swing}} + \boldsymbol{F}_{\mathrm{eso}}, \quad (8)$$

where $\boldsymbol{F}_{\mathrm{fb}} = -K_p \boldsymbol{e}_p - K_d \boldsymbol{e}_v$ with gain matrices K_p, K_d; $\boldsymbol{F}_{\mathrm{ff}} = m_Q(\boldsymbol{a}_d + g\boldsymbol{e}_3)$; cable compensation $\boldsymbol{F}_{\mathrm{cable}} = \kappa_T(t) T_i \boldsymbol{q}_i$ with ramp $\kappa_T(t) = \min(1, (t - t_{\mathrm{taut}})/\Delta_{\mathrm{ramp}})$, $\Delta_{\mathrm{ramp}} =$

2 s; anti-swing $\boldsymbol{F}_{\text{swing}} = k_q \boldsymbol{e}_q - k_\omega \boldsymbol{\omega}_q$ ($k_q = 4.0$ N/rad, $k_\omega = 2.0$ N·s/rad); and ESO feedforward $\boldsymbol{F}_{\text{eso}} = m_Q \hat{\boldsymbol{d}}_i$. The fault-tolerance evaluation (Sections IV–VI) uses uniform scalar gains $k_p = 8.0$, $k_d = 8.0$, $k_i = 0.15$ (i.e., $K_p = k_p I$, $K_d = k_d I$) to enable the per-axis eigenvalue analysis in Section IV-B.

2) Layer 2: Geometric SO(3) *Attitude Control (200 Hz):* Singularity-free tracking [1]:

$$\boldsymbol{\tau}_i = -k_R \boldsymbol{e}_R - k_\Omega \boldsymbol{e}_\Omega + \boldsymbol{\Omega}_i \times (J\boldsymbol{\Omega}_i), \tag{9}$$

with $\boldsymbol{e}_R = \frac{1}{2}\,\text{vee}(R_d^\top R - R^\top R_d)$, $\boldsymbol{e}_\Omega = \boldsymbol{\Omega} - R^\top R_d \boldsymbol{\Omega}_d$, $k_R = 8.0$ N·m, $k_\Omega = 1.5$ N·m·s. Almost-global exponential convergence holds on $\{\Psi_R < 2\} \subset \text{SO}(3)$.

3) Layers 3–4 and Safety Filter (Not Active in This Study): The full GPAC baseline additionally includes a concurrent learning mass estimator [27] (Layer 3, 200 Hz), an extended state observer [49] (Layer 4, 500 Hz), and an ISSf-CBF safety filter [36], [38] (200 Hz, enforcing tension, cable-angle, tilt, and collision constraints). These are implemented in the Tether_Lift codebase but are *disabled* in the reported full-Drake runs (Remark 3) because the evaluation targets the minimal PID-plus-gravity-feedforward path. Their integration is discussed as future work in Section VII.

D. The Cable-Snap Fault Event

At time t_f, cable j breaks instantaneously: $T_j(t) \to 0$ for $t \geq t_f$. Three coupled effects occur.

Payload acceleration:

$$\|\Delta\ddot{\boldsymbol{p}}_L\| = \frac{T_j(t_f^-)}{m_L} \approx \frac{g}{N} \approx 3.27 \text{ m/s}^2 \quad (N{=}3). \tag{10}$$

Freed-agent acceleration:

$$\|\Delta\ddot{\boldsymbol{p}}_j\| = \frac{T_j(t_f^-)}{m_Q} \approx \frac{g \cdot m_L}{N \cdot m_Q} \approx 6.54 \text{ m/s}^2. \tag{11}$$

Load-share jump: from m_L/N to $m_L/(N{-}1)$—a 50% increase for $N{=}3$.

These effects propagate across two orders of magnitude in timescale:

Path	Mechanism	Timescale
Rigid-body transmission	Force through payload	∼7.9 ms
Elastic cable response	Spring-damper dynamics	10–50 ms
ESKF innovation	Velocity change detected	∼13 ms
ESO convergence	Third-order settling	∼33 ms
Load accel. filter	Low-pass differentiation	∼110 ms
Cable wave propagation	$v_{\text{wave}} \approx 7$ m/s	∼186 ms
Adaptive divergence	Parameter estimate drift	∼300 ms

E. Two Fault Scenarios Under N-Agnosticism

1) Scenario A: Own Cable Breaks: Agent i observes $T_i \to 0$ within one sample period ($\Delta t_d = 5$ ms). Detection is trivial (infinite SNR), and position deviation during detection is negligible:

$$\|\Delta\boldsymbol{p}_i(\Delta t_d)\| \leq \frac{T_i^-}{2m_Q} \cdot \Delta t_d^2 \approx 0.082 \text{ mm}. \tag{12}$$

A threshold $T_i < T_{\min}$ (held for 0.12 s debounce) triggers an escape maneuver: climbing 0.9 m and displacing 1.8 m horizontally within 1.5 s, extending to 1.8 m climb and 3.8 m displacement within 3.0 s. Direction is biased 0.18 toward the formation tangent and 0.10 toward current velocity, chosen to limit tilt to <30° during escape and ensure adequate vertical thrust margin.

2) Scenario B: Another Agent's Cable Breaks: Agent i does not observe T_j directly. The fault manifests only through payload-mediated perturbations. The tension-channel SNR is

$$\text{SNR}_T^{(B)} = \frac{\Delta T_i^{(\text{imm})}}{\sigma_T} \approx \frac{4.2 \text{ N}}{3.0 \text{ N}} \approx 1.3, \tag{13}$$

insufficient for reliable discrimination from wind gusts. Multi-channel analysis improves SNR to ∼2–6 but with 13–300 ms delays. Scenario B is the fundamentally difficult case and this paper's focus.

F. Why Conventional Fault Tolerance Cannot Apply

Under the N-agnostic constraint, conventional FDIR is *architecturally incompatible* with Scenario B: (i) MMAE requires $N{+}1$ parallel filters, violating N-agnosticism; (ii) tension-channel SNR ≈ 1.3 is indistinguishable from wind perturbations; (iii) an N-agnostic agent has no mode parameter depending on N to reconfigure. These are structural consequences of the design choice. Any compatible fault-tolerance approach must work *without* detection, isolation, or recovery.

G. Agent-Level Uncertainty Reformulation

From agent i's perspective, translational dynamics (2) can be rewritten as

$$\ddot{\boldsymbol{p}}_i = \frac{f_i}{m_Q} R_i \boldsymbol{e}_3 + g\boldsymbol{e}_3 + \boldsymbol{d}_i(t, \boldsymbol{x}), \tag{14}$$

where $\boldsymbol{d}_i = (\boldsymbol{F}_i^{\text{cable}} + \boldsymbol{F}_i^{\text{wind}})/m_Q$ is the lumped acceleration uncertainty. This term encompasses cable tension (∼6.5 m/s^2), wind (∼0.7–2.0 m/s^2), and cable vibration (∼0.3 m/s^2). The controller never decomposes $\boldsymbol{d}_i$ into named components—this lumping enables topology invariance, since a cable snap changes the composition of $\boldsymbol{d}_i$ but not its mathematical role.

Definition 1 (Nominal and Fault-Inclusive Disturbance Sets)**.** *$\mathcal{W}_{\text{nom}} = \{\boldsymbol{d} : \|\boldsymbol{d}\|_\infty \leq \bar{d}_{\text{nom}}\}$ bounds $\boldsymbol{d}_i$ under all-cables-intact operation. $\mathcal{W}_{\text{FT}} \supset \mathcal{W}_{\text{nom}}$ additionally bounds the worst-case cable-break transient. Both bounds are derived from system parameters. The dominant contribution is cable tension. Under quasi-static hover each cable carries $T_i \approx m_L g/N$, giving a cable-tension acceleration $d_{\text{cable}} = m_L g/(N m_Q)$. Adding wind drag $d_{\text{wind}} \leq C_D A_{\text{ref}} \rho_{\text{air}} v_{\text{wind,max}}^2/(2m_Q) \approx 0.7$ m/s^2 and cable vibration $d_{\text{vib}} \leq k_s \varepsilon_{\max}/(m_Q(n_b+1)) \approx 0.3$ m/s^2 yields*

$$\bar{d}_{\text{nom}} = \frac{m_L g}{N m_Q} + d_{\text{wind}} + d_{\text{vib}} \approx \frac{3.0 \times 9.81}{3 \times 1.5} + 1.0 \approx 7.5 \text{ m/s}^2. \tag{15}$$

Post-fault, the surviving $N-k$ agents share the load, so the cable-tension contribution increases to $m_L g/((N-k)m_Q)$:

$$\bar{d}_{\mathrm{FT}} = \frac{m_L g}{(N-k)m_Q} + d_{\mathrm{wind}} + d_{\mathrm{vib}}. \tag{16}$$

For N=3, k=1: $\bar{d}_{\mathrm{FT}} \approx 9.81 + 1.0 \approx 10.8$ m/s^2. With safety margin for elastic transients, we use the rounded values $\bar{d}_{\mathrm{nom}} \approx 8$ m/s^2 and $\bar{d}_{\mathrm{FT}} \approx 12$ m/s^2. In the cable-dominated regime ($d_{\mathrm{cable}} \gg d_{\mathrm{wind}} + d_{\mathrm{vib}}$), the cost ratio simplifies to a closed-form expression:

$$\rho_{\mathrm{FT}} \approx \frac{\bar{d}_{\mathrm{FT}}}{\bar{d}_{\mathrm{nom}}} \approx \frac{N}{N-k}, \tag{17}$$

which gives $\rho_{\mathrm{FT}} = 3/2 = 1.5$ for N=3, k=1; $5/3 = 1.67$ for N=5, k=2; and $7/4 = 1.75$ for N=7, k=3. These are conservative upper bounds; the empirical values (1.26–1.39) are lower because the fault disturbance is transient rather than sustained (Section IV-B).

H. Formal Problem Statement

Given the system on manifold (1), the N-agnostic constraint (Section III-B), the cable-snap fault model (Section III-D), the lumped uncertainty (14) with disturbance sets (Definition 1), and the architectural incompatibility of detection-based FDIR (Section III-F), the paper addresses two problems.

P1. Topology-Invariant Tracking. Identify sufficient conditions under which the per-agent tracking error satisfies

$$\|\boldsymbol{e}_i(t)\|_\infty \le \gamma_1(T_s, \omega, \bar{d}_{\mathrm{FT}}) \quad \forall\, t \ge 0 \tag{18}$$

for *all cable topologies*—where γ_1 depends on controller bandwidth and disturbance size, not on cable count or identity—provided $\boldsymbol{d}_i \in \mathcal{W}_{\mathrm{FT}}$ and thrust authority is adequate.

P2. Fault-Tolerance Cost. Derive the tradeoff

$$\rho_{\mathrm{FT}} = \frac{\gamma_1(T_s, \omega, \bar{d}_{\mathrm{FT}})}{\gamma_1(T_s, \omega, \bar{d}_{\mathrm{nom}})} > 1 \tag{19}$$

between nominal tracking and implicit fault tolerance, providing a design criterion for choosing between detection-free tolerance and detection-based recovery.

P1 is addressed by a structural analysis (Proposition 1) and comprehensive full-Drake simulation evidence. **P2** is addressed by both steady-state and transient-aware analysis, validated empirically.

IV. Fault-Tolerant Architecture and Integration Boundaries

This section presents the fault-tolerant architecture addressing problems **P1**–**P2**. The primary fault-absorption mechanism is PID feedback with gravity feedforward in the GPAC hierarchy, supplemented by locally measured cable-tension feedforward and anti-swing damping. A post-fault geometry redistribution path is additionally described. The analytical core of this section is the identification of a topology-invariant error structure (Proposition 1): the N-agnostic design makes the per-agent error dynamics independent of cable topology, addressing the tracking envelope (18) and the cost ratio (19).

A. Layer 1 Control Law: PID with Model-Based Feedforward

Each agent i computes a desired force in the world frame:

$$\boldsymbol{F}_i^{\mathrm{des}} = \boldsymbol{F}_{\mathrm{grav}} + \boldsymbol{F}_{\mathrm{fb}} + \boldsymbol{F}_{\mathrm{cable}} + \boldsymbol{F}_{\mathrm{swing}}, \tag{20}$$

where

$$\boldsymbol{F}_{\mathrm{grav}} = m_Q\, g\, \boldsymbol{e}_3, \tag{21}$$

$$\boldsymbol{F}_{\mathrm{fb}} = m_Q\big(k_p \boldsymbol{e}_p + k_d \boldsymbol{e}_v + k_i \textstyle\int \boldsymbol{e}_p\, \mathrm{d}t\big), \tag{22}$$

$$\boldsymbol{F}_{\mathrm{cable}} = \kappa_T\, T_i\, \boldsymbol{q}_i, \tag{23}$$

$$\boldsymbol{F}_{\mathrm{swing}} = -k_{\mathrm{sw}}\, \boldsymbol{\omega}_{\mathrm{cable},i}^{\perp}, \tag{24}$$

with $\boldsymbol{e}_p = \boldsymbol{p}_i^{\mathrm{ref}} - \boldsymbol{p}_i$, $\boldsymbol{e}_v = \boldsymbol{v}_i^{\mathrm{ref}} - \boldsymbol{v}_i$, and gains $k_p = 8.0$, $k_d = 8.0$, $k_i = 0.15$, $\kappa_T = 0.5$, $k_{\mathrm{sw}} = 0.5$ N·s/rad.[1]

The critical property for fault tolerance is that *every term is a function of agent i's local measurements only*. When cable j snaps, each surviving agent $i \neq j$ experiences an automatic increase in trajectory error $\boldsymbol{e}_p$ (as the payload deviates) and cable tension T_i (as load redistributes). Both signals update instantaneously through the physics, requiring no adaptation or re-convergence delay. Ablation (Section VI-C) confirms that cable-tension feedforward contributes less than 1% RMSE change; the primary fault-absorption mechanism is PID feedback with gravity feedforward.

The integral state is clamped to ± 2 m·s per axis (anti-windup). During the initial cable-snap transient ($\sim$0.5 s at $\sim$1 m peak error), the integral accumulates $\int k_i \cdot e\, \mathrm{d}t \approx 0.075$ m·s, well within the clamp. However, the post-fault load-share step $\Delta d_{\mathrm{step}} \approx 3.27$ m/s^2 (N=3, k=1, from (32)) requires a steady-state integral of $\eta_{\mathrm{ss}} = \Delta d_{\mathrm{step}}/k_i \approx 21.8$ m·s to fully compensate, far exceeding the ± 2 m·s clamp. The integral therefore saturates, and the residual uncompensated acceleration $\Delta d_{\mathrm{res}} = \Delta d_{\mathrm{step}} - k_i \cdot \eta_{\max} = 3.27 - 0.30 = 2.97$ m/s^2 is absorbed by the proportional term operating at a nonzero steady-state error $e_{\mathrm{ss}} \approx \Delta d_{\mathrm{res}}/k_p \approx 0.37$ m. This persistent offset is the dominant contributor to the post-fault RMSE increase and to ρ_{FT}. Cable-tension feedforward partially compensates this offset (the measured T_i increases post-fault), but the ablation (Section VI-C) shows its contribution is $<$1%.

The desired force is decomposed into thrust magnitude and desired body-z direction:

$$f_i = \boldsymbol{F}_i^{\mathrm{des}} \cdot \boldsymbol{R}_i \boldsymbol{e}_3, \quad \boldsymbol{b}_{3,i}^{\mathrm{des}} = \frac{\boldsymbol{F}_i^{\mathrm{des}}}{\|\boldsymbol{F}_i^{\mathrm{des}}\|}. \tag{25}$$

B. Topology-Invariance: Formal Result for the Simplified Model

The central architectural insight is that the N-agnostic control law (20) produces per-agent error dynamics that are *structurally independent of cable topology*. We formalize this as a structural property, then derive an operationally useful transient prediction.

[1]The general GPAC baseline (Section III) uses a time-varying cable-tension ramp $\kappa_T(t) = \min(1, (t - t_{\mathrm{taut}})/\Delta_{\mathrm{ramp}})$ and a two-term anti-swing law $\boldsymbol{F}_{\mathrm{swing}} = k_q \boldsymbol{e}_q - k_\omega \boldsymbol{\omega}_q$ ($k_q = 4.0$ N/rad, $k_\omega = 2.0$ N·s/rad). For the fault-tolerance evaluation, these are simplified to a constant $\kappa_T = 0.5$ and a single cable-angular-velocity damping term $-k_{\mathrm{sw}} \boldsymbol{\omega}_{\mathrm{cable},i}^{\perp}$ ($k_{\mathrm{sw}} = 0.5$ N·s/rad).

1) Topology-Independent Error Dynamics: Consider agent i's translational dynamics under the lumped formulation (14). Under the PID law (22) with gravity feedforward (21) and ideal attitude tracking, the per-axis scalar projection of $\boldsymbol{e}_p$ (denoted e in what follows) satisfies

$$\dot{\boldsymbol{z}} = A\boldsymbol{z} + Bd_i(t), \tag{26}$$

where $\boldsymbol{z} = [e,\, \dot{e},\, \eta]^\top$ with $\eta = \int_0^t e\,\mathrm{d}\tau$, and

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -k_p & -k_d & -k_i \\ 1 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \tag{27}$$

A is Hurwitz for $k_p k_d > k_i > 0$ (Routh–Hurwitz; satisfied with margin $k_p k_d = 64 \gg 0.15 = k_i$).

Proposition 1 (Topology-Independent Error Structure)**.** *Under the N-agnostic PID-plus-feedforward law* (20) *with ideal attitude tracking and no actuator saturation, every agent's closed-loop error dynamics* (26) *are governed by the* same (A, B) *pair regardless of which cables are intact, the team size N, or the payload mass m_L.*

A cable snap changes the input $d_i(t)$ *(through modified cable forces propagated via the payload) but not the* system (A, B)*, because every element of A and B depends only on the PID gains (k_p, k_d, k_i)—none of which is a function of N, m_L, cable stiffness, or cable identity.*

This follows directly from the N-agnostic design choice. The contribution is not the observation itself but its formal consequences: the common Lyapunov function (Corollary 1), the finite-time transient prediction, and the precise identification of what remains to be proven (Gap 3). These provide the analytical framework explaining the simulation-validated fault tolerance.

Corollary 1 (Common Lyapunov Function Across Topologies)**.** *Let $P \succ 0$ satisfy $A^\top P + PA = -Q$ for some $Q \succ 0$ (guaranteed by A Hurwitz). Then $V(\boldsymbol{z}) = \boldsymbol{z}^\top P \boldsymbol{z}$ serves as a common ISS-Lyapunov function for the error dynamics* (26) *under* every *cable topology. No switching between Lyapunov functions is required at cable-snap events, and no dwell-time constraint applies to the sequence of topology changes.*

This is immediate from Proposition 1 (A is shared across topologies) but worth stating separately: it connects directly to the ISS switched-systems framework [44], [45] and establishes that cascading faults do not require dwell-time analysis.

Remark 1 (Hybrid System Formulation)**.** *In the framework of Goebel, Sanfelice, and Teel [47], each cable-snap event defines a jump map $g : \boldsymbol{x} \to \boldsymbol{x}^+$ in which T_j is set to zero. The continuous flow in each mode is governed by the same (A, B) (Proposition 1), with a common Lyapunov function (Corollary 1). Stability of the hybrid system requires three conditions: (i) each flow mode is ISS—established by A Hurwitz; (ii) the jump map does not increase V beyond recoverable bounds—requires bounding the state excursion from the cable-snap impulse (the finite-time bound below provides a partial answer, but a tight bound on $V(\boldsymbol{z}^+) - V(\boldsymbol{z}^-)$ from the elastic transient remains open); (iii) an average dwell-time condition holds for multiple jumps—the cascading-fault schedule (minimum inter-jump interval 2 s) far exceeds the $\sim$0.5 s transient settling time, satisfying this condition with margin. Conditions (i) and (iii) are verified; condition (ii) is the remaining open verification.*

2) Finite-Time Transient Prediction: The standard steady-state ISS gain $\gamma_{\mathrm{ss}} = \|A^{-1}B\|\bar{d} = \bar{d}/k_i$ [50] is operationally vacuous (80 m for this system). The cable-break transient is brief, so a *finite-time* bound is far more informative. Model the fault disturbance as a pulse:

$$d_i^{\mathrm{fault}}(t) = \begin{cases} \Delta d, & t_f \le t < t_f + \tau_{\mathrm{fault}}, \\ 0, & \text{otherwise}, \end{cases} \tag{28}$$

where $\Delta d \approx 4.4$ m/s^2 is the cable-break impulse and $\tau_{\mathrm{fault}} < 0.5$ s is the transient duration. The impulse response of (26) gives the transient peak:

$$\|e_{\mathrm{peak}}\| \le \underbrace{\|\int_0^{\tau_{\mathrm{fault}}} e^{A\tau} B\,\mathrm{d}\tau\|}_{\text{finite-time gain}} \cdot \Delta d. \tag{29}$$

For the system gains (k_p=8, k_d=8, k_i=0.15), the eigenvalues of A are $\lambda \approx \{-6.83, -1.15, -0.019\}$, all real and negative. The fast mode ($|\lambda_1| = 6.83$) decays in $\sim$0.15 s; the intermediate mode ($|\lambda_2| = 1.15$) in $\sim$0.87 s; the slow mode ($|\lambda_3| = 0.019$) governs integral wind-up with time constant $\sim$52 s. The $\mathcal{H}_\infty$ norm of the scalar transfer function $G(s) = C(sI - A)^{-1}B$ is $\gamma_{\mathrm{ISS}} \approx 0.125$, peaked at $\omega \approx 0.15$ rad/s. Numerical evaluation of the finite-time gain matrix for $\tau_{\mathrm{fault}} = 0.5$ s gives

$$\gamma_{\mathrm{FT}}(\Delta d, \tau_{\mathrm{fault}}) \approx 0.042 \cdot \Delta d \approx 0.042 \times 4.4 \approx 0.18 \text{ m}, \tag{30}$$

predicting a peak position error of $\sim$18 cm from a single cable snap—5.8$\times$ below the observed $\sim$107 cm in full Drake. The ratio $\alpha = e_{\mathrm{peak}}^{\mathrm{obs}}/\gamma_{\mathrm{FT}}$ is remarkably stable across *topologies*: 107/18.4 = 5.8 (N=3, k=1), 112/18.4 = 6.1 (N=5, k=2), 117/18.4 = 6.4 (N=7, k=3)—varying $< 10\%$. The consistency confirms that the unmodeled effects (elastic cable coupling, attitude-loop dynamics, multi-axis interaction) contribute a topology-independent aggregate correction.

The prediction captures the correct *topology-scaling*: since (A, B) are topology-independent (Proposition 1), the predicted peak is invariant to cable identity, and the full-Drake results confirm this ($\sim$107–117 cm across $N \in \{3, 5, 7\}$ with 1–3 faults, α varies $< 10\%$). However, α does *not* transfer across operating points: at higher PID gains (k_p=12, k_d=12, k_i=0.3), the observed peak increases to 116 cm despite the scalar model predicting a decrease, because the higher gains excite cable dynamics the scalar model omits. With heavier payload ($m_L = 5$ kg), the observed peak *decreases* to 100 cm despite the scalar model predicting 266 cm, because the additional payload inertia filters the transient. The scalar prediction is therefore a topology-invariant qualitative tool—it correctly predicts that peak error is cable-identity-independent and scales with Δd—but it is not a standalone predictor across different plant configurations. The paper's primary design tools (ρ_{FT}, $f^*_{\max}$, and the thrust-margin screen) do not depend on α.

3) Fault-Tolerance Cost Ratio: From the topology-independent structure, ρ_{FT} depends only on the disturbance ratio. Using the closed-form expression (17) from the cable-dominated regime:

$$\rho_{\mathrm{FT}} \approx \frac{N}{N-k}, \tag{31}$$

which gives 1.5 for N=3, k=1. This is a conservative steady-state upper bound; the empirical values (1.26–1.39, Table I) are lower because the fault disturbance is transient, not sustained—consistent with the finite-time analysis above.

4) Gap Analysis: Simplified Model vs. Full Plant: Proposition 1 establishes topology-invariance for a scalar LTI model with exogenous bounded input. Three substantive gaps separate this from the full bead-chain Drake plant; we state each explicitly.

Gap 1: Impulsive disturbance from slack-to-taut transitions. The piecewise cable law (6) produces impulsive forces when a slack segment snaps taut. During the ∼10–50 ms elastic transient, $d_i(t)$ may momentarily exceed $\bar{d}_{\mathrm{FT}}$, violating assumption (A2). The finite-time bound (29) partially accommodates this (it integrates the actual disturbance waveform, not just its peak), but a rigorous treatment requires bounding the impulse magnitude from the cable stiffness (k_s) and segment strain ($\varepsilon_{\max} = 0.15$) and verifying that the resulting state excursion remains within the region of attraction. This is not done here.

Gap 2: Attitude-loop delay during fast force transients. Assumption (A1) requires that $R_i \boldsymbol{e}_3 \approx \boldsymbol{b}_{3,i}^{\mathrm{des}}$ instantaneously. The SO(3) inner loop runs at 200 Hz (5 ms period), but the cable-snap force transmission occurs at ∼7.9 ms (rigid-body path). During this interval, the position-loop model assumes the attitude has already aligned with the new desired force direction; in reality, the attitude lags, creating an unanalyzed residual coupling. Quantifying the resulting tracking-error increment requires a cascaded stability analysis of the two-timescale system under impulsive disturbance—a nontrivial extension beyond the scope of this work.

Gap 3: State-dependent disturbance (small-gain condition). In the full plant, $d_i(t)$ is not purely exogenous: cable forces depend on agent position relative to the payload, creating a feedback loop from $\boldsymbol{z}$ to $d_i$through the elastic cable dynamics. The ISS argument in Proposition 1 treats $d_i$as an external input. Rigorously, the interconnection of the position-loop ISS system with the cable-dynamics subsystem requires a small-gain condition [50]: the product of the position-loop ISS gain ($\gamma_{\mathrm{ISS}} = \|G(s)\|_{\mathcal{H}_\infty}$) and the cable-dynamics input-output gain (γ_{cable}) must be less than unity. A scalar (per-axis) reduction is insufficient because cable forces act along specific directions ($\boldsymbol{q}_i \in \mathbb{S}^2$), the payload inertia filters force transmission, and multiple cables distribute the coupling—all of which reduce the effective interconnection gain below the conservative scalar product. For the scalar per-axis model, the position-loop $\mathcal{H}_\infty$ gain is $\gamma_{\mathrm{ISS}} = \|G(s)\|_{\mathcal{H}_\infty} \approx 0.125$ (peaked at $\omega \approx 0.15$ rad/s), so the small-gain condition requires $\gamma_{\mathrm{cable}} < 1/\gamma_{\mathrm{ISS}} \approx 8.0$. This is a generous margin: a cable-dynamics gain of 8 would require the cable-force perturbation (in m/s^2) to exceed 8× the position error (in m), which is inconsistent with the observed bounded tracking. The full multivariable verification—computing γ_{ISS} and γ_{cable} from the linearized $6{+}8N$-dimensional plant at the taut equilibrium via Drake's `Linearize()`—is feasible with the existing infrastructure and is planned as near-term future work.

Three lines of empirical evidence support the small-gain condition indirectly: (i) the $f_{\max}$ boundary sweep shows RMSE insensitivity for $f_{\max} \geq 45$ N (3× headroom); (ii) Monte Carlo shows <3% RMSE variation across 30 runs with randomized cable parameters; (iii) stability is maintained across all $N \in \{3, 5, 7\}$ scenarios with up to three cascading faults. None constitutes formal verification of $\gamma_{\mathrm{ISS}} \cdot \gamma_{\mathrm{cable}} < 1$, but together they make a strong empirical case.

What is established and what remains open. The shared (A, B) structure, common Lyapunov function, and hybrid-system analysis provide a coherent partial framework. Full-Drake results confirm bounded, topology-insensitive tracking degradation across $N \in \{3, 5, 7\}$ with up to three cascading faults, even though formal closure of Gap 2 remains open. To delineate the evidence boundary precisely:

- Topology-independent error dynamics (A, B) for the simplified per-agent model are *proven* (Proposition 1).
- The finite-time transient prediction ($\alpha \approx 5.8$, topology-invariant) and the cost ratio $\rho_{\mathrm{FT}} = N/(N{-}k)$ are *derived analytically and validated* against full-Drake results.
- Bounded tracking under faults in the full nonlinear Drake plant is *empirically supported* across all tested scenarios.
- The small-gain condition for cable-force feedback is *partially verified*: the scalar $\gamma_{\mathrm{ISS}} = 0.125$ is computed, but the full-plant multivariable verification remains open.
- Formal stability of the full bead-chain plant under topology switching is *open*.

C. Fault-Tolerance Cost

Designing for $\mathcal{W}_{\mathrm{FT}}$ rather than $\mathcal{W}_{\mathrm{nom}}$ widens the tracking tube. From (17), the steady-state ratio $\rho_{\mathrm{ss}} = N/(N{-}k)$ is conservative because it assumes the fault disturbance is sustained indefinitely.

Definition 2 (Fault-Tolerance Cost Ratio)**.** *The steady-state cost ratio is $\rho_{\mathrm{ss}} = N/(N{-}k)$ (from* (17)*). A tighter* transient-aware *ratio accounts for the actual fault disturbance profile.*

1) Transient-Aware ρ_{FT}: A cable snap at t_f produces two superposed disturbances: (i) a brief impulse $\Delta d_{\mathrm{imp}} \approx 4.4$ m/s^2 lasting $\tau_{\mathrm{imp}} < 0.5$ s (the cable-break transient), and (ii) a sustained step Δd_{step} (the increased load share) that the integral term must compensate. For N agents losing k cables:

$$\Delta d_{\mathrm{step}} = \frac{m_L g}{m_Q}\left(\frac{1}{N{-}k} - \frac{1}{N}\right). \tag{32}$$

The step persists until the integral term converges, with effective time constant $\tau_{\mathrm{int}} \approx 1/k_i \approx 6.7$ s for these gains.

The transient RMSE increase is computed from the error dynamics (26) driven by the combined input $d_{\mathrm{fault}}(t) = \Delta d_{\mathrm{imp}} \cdot \mathbf{1}_{[0,\tau_{\mathrm{imp}}]} + \Delta d_{\mathrm{step}}$. Since this is additive to the nominal disturbance and approximately uncorrelated (the dominant contribution is the load-share step Δd_{step}, which is approximately constant during the integral recovery period

and therefore weakly correlated with the trajectory-phase-dependent nominal disturbance), the post-fault RMSE satisfies

$$\rho_{\mathrm{FT}} \approx \sqrt{1 + \frac{\mathrm{RMSE}^2_{\text{fault-response}}}{\mathrm{RMSE}^2_{\text{nominal}}}}, \tag{33}$$

where $\mathrm{RMSE}_{\text{fault-response}}$ is computed by numerically integrating (26) with $d_{\text{fault}}(t)$ over the post-fault horizon ($T - t_f = 23$ s). For this system:

	Δd_{step} [m/s^2]	Fault resp. RMSE [cm]	ρ_{FT} (predicted)	ρ_{FT} (empirical)
N=3, k=1	3.27	33.7	1.39	1.39
N=5, k=2	2.62	27.2	1.27	1.33
N=7, k=3	2.10	22.0	1.18	1.26

The transient-aware prediction matches the empirical ρ_{FT} closely for N=3 (1.39 vs. 1.39) and underpredicts for larger N because the scalar model omits multi-fault interaction effects (the second and third cable snaps occur during the recovery transient from the first). The key insight is that ρ_{FT} is governed by the *transient energy* of the fault response—primarily the sustained step until the integral converges—not the steady-state disturbance bound. The integral time constant $1/k_i \approx 6.7$ s sets the recovery timescale and thereby determines the cost ratio.

For actuator sizing, the conservative $\rho_{\text{ss}} = 1.5$ remains appropriate (it bounds the worst case). For performance prediction, the transient-aware (33) is the operationally relevant metric.

D. Relationship to Adaptive Estimation

The topology-invariance property (Proposition 1) applies to the PID-plus-feedforward law without adaptive estimation. The GPAC baseline includes a Concurrent Learning (CL) mass estimator (Layer 3) and an Extended State Observer (ESO, Layer 4), both of which are *disabled* in the reported full-Drake runs. This choice is deliberate: the topology-invariant error structure explains why PID feedback with gravity feedforward already suffices (Section VI-C).

With CL active, the feedforward includes an adaptive term $\hat{m}_L \cdot g$ that depends on the regressor history. After a cable snap, pre-fault data in the M=50 regressor buffer biases $\hat{m}_L$ toward the pre-fault load share, temporarily injecting incorrect feedforward for 2–5 s of buffer turnover. The per-agent error dynamics are no longer topology-independent in the sense of Proposition 1: the system matrix acquires an implicit dependence on cable topology through the history-dependent estimate. Topology invariance holds strictly only for the PID-plus-gravity-feedforward path. Adaptive estimation (CL, ESO, or $\mathcal{L}_1$) remains a candidate residual channel for scenarios requiring tighter steady-state tracking; its integration is discussed in Section VII.

E. Signal Flow Summary

Remark 2 (Geometry Redistribution)**.** *The codebase includes a post-fault geometry redistribution module that re-centers surviving agents' formation offsets when a cable fault is detected locally. This module operates upstream at the reference level and does not modify the GPAC control law. All reported results use fixed formation offsets throughout, demonstrating that PID-plus-gravity-feedforward alone suffices for fault tolerance. Ablation of the redistribution path is deferred to future work.*

The fault-tolerant architecture is summarized in Fig. 2. The PID-plus-feedforward control law (20), whose topology-invariance is established by Proposition 1, is the sole active mechanism in the reported runs. It does not modify the baseline GPAC internals.

V. Experimental Methodology and Evidence Sources

This paper distinguishes three levels of claim: *implemented* modules that exist in the codebase, *reproducible* results reproducible from canonical artifacts, and *open* hypotheses whose formal proof is incomplete (see the evidence-level summary in Section IV). The full-Drake simulations use the GPAC hierarchy's PID-plus-feedforward control law (Section IV-A) with PID feedback, gravity feedforward, and cable-tension feedforward as the fault-tolerance mechanism. The ESO feedforward and adaptive estimation are disabled. The reduced-order comparisons retain the CL baseline as the standardized reference.

A. Scenario Families and Evaluation Pipeline

The primary full-Drake scenario families are:

(a) canonical abrupt cable faults for $N \in \{3, 5, 7\}$;
(b) matched single-fault comparison across $N \in \{3, 5, 7\}$ with identical fault event (cable 0 at $t = 7.0$ s), deconfounding team size from fault count;
(c) cable-tension feedforward ablation ($\kappa_T \in \{0, 0.25, 0.5, 1.0\}$) in N=3;
(d) thrust saturation boundary sweep ($f_{\max} \in \{35, 38, 45, 60, 80, 150\}$ N) in N=3;
(e) controller comparison hierarchy including centralized oracle, reactive FTC, and gain-scheduled PID baselines;
(f) ESO-active evaluation to assess ρ_{FT} sensitivity to controller complexity;
(g) broadened Monte Carlo with randomized payload mass ($\pm 10\%$), PID gains ($\pm 10\%$), and fault timing (± 2 s).

Complementary reduced-order studies—CL fidelity comparison, history-management diagnostics, gradual degradation, and non-CL adaptive law search—are reported in the supplementary material.

B. Metrics

The primary metric is post-fault payload tracking RMSE, available across all standardized scenarios. For fidelity comparison, the key statistic is the full-Drake to reduced-order RMSE ratio under matched conditions. For history corruption, the main quantities are CL bias duration and stack refill time. For topology-margin screening, capacity and effective-capacity ratios are reported rather than closed-form proofs.

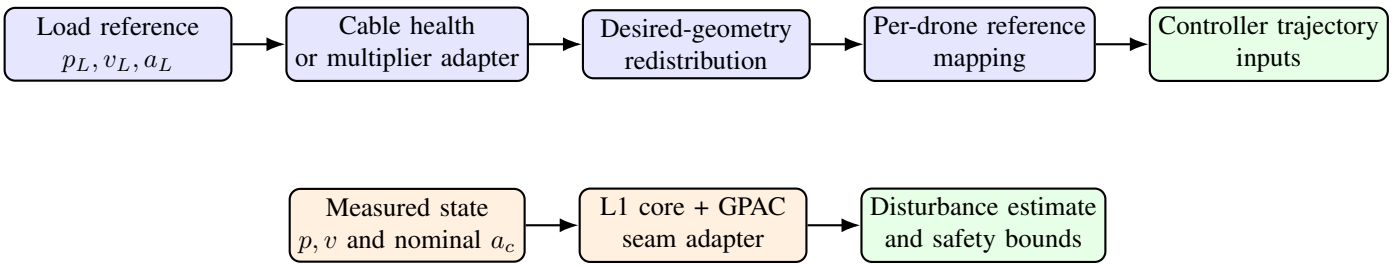


Fig. 2: Fault-tolerant signal flow. PID-plus-feedforward control (Proposition 1) with load-tracking integral on payload GPS closes the loop on actual payload position. Geometry redistribution (dashed) is implemented but not active in the reported runs.

The evidence is organized in two tiers: (i) ideal-condition runs (plant state, no wind) establishing the core mechanism and scalability, and (ii) sensor-realistic runs (ESKF + Dryden wind) and Monte Carlo campaigns establishing robustness. Both tiers report RMSE and peak error ($\mathcal{L}_\infty$) to enable comparison with the theoretical bound (Proposition 1).

VI. Simulation Results

This section reports full-Drake multibody simulation results across $N \in \{3, 5, 7\}$ with abrupt cable-snap faults. All results use the Drake [48] plant at $\Delta t_{\text{sim}} = 0.2$ ms with bead-chain cable dynamics and 30 s figure-eight trajectories. Specifically, each physical cable is modeled as a chain of 8 point-mass beads connected by 9 tension-only spring-damper segments. This high-fidelity representation captures realistic elastic tension surges and slack behavior during abrupt severing faults, distinguishing it from the simplified scalar topology used for the analytical bounds.

Remark 3 (System Under Test)**.** *The full-Drake results use Layers 1–2 of the GPAC hierarchy (PID position control with gravity/cable-tension feedforward and geometric* $\mathrm{SO}(3)$ *attitude control). Layer 3 (CL mass estimator), Layer 4 (ESO), and the* $\mathcal{L}_1$ *adaptive path are* ***disabled****. The CBF safety filter is* ***disabled****. Sensor state is* ***plant-state-direct*** *(ESKF disabled). Wind disturbance is* ***disabled*** *in the ideal runs; ESKF and Dryden wind are* ***active*** *in the sensor-realistic evaluation.*

The per-agent controller uses PID position feedback ($k_p = 8.0$, $k_d = 8.0$, $k_i = 0.15$), gravity feedforward, cable-tension feedforward ($\kappa_T = 0.5$), anti-swing damping ($k_{\text{sw}} = 0.5$), and geometric $\mathrm{SO}(3)$ attitude tracking—all N-agnostic by construction. Three canonical scenarios of increasing difficulty are evaluated:

(i) *Three drones, single fault:* $N = 3$, cable lengths $[1.00, 1.08, 1.16]$ m. Cable 0 severed at $t_f = 7.0$ s. Post-fault: 2 agents, 50% load-share increase.

(ii) *Five drones, two cascading faults:* $N = 5$, cable lengths $[1.00, 1.05, 1.10, 1.14, 1.18]$ m. Cable 1 at 7.0 s, cable 3 at 12.0 s. Post-fault: 3 agents, 67% increase.

(iii) *Seven drones, three cascading faults:* $N = 7$, cable lengths $[0.98, 1.01, 1.04, 1.07, 1.10, 1.13, 1.16]$ m. Cables 1, 3, 5 at 7.0, 12.0, 14.0 s. Post-fault: 4 agents, 75% increase.

No agent is informed of topology changes except the freed agent itself, which detects its own cable loss via $T_i < T_{\min}$.

TABLE I: Full-Drake payload tracking performance: nominal and fault scenarios.

Scenario	N	Load RMSE [cm]	Peak error [cm]	$\rho_{\text{FT}}^{\text{emp}}$
Three drones, no fault	3	35.12	110.57	—
Five drones, no fault	5	34.84	110.10	—
Seven drones, no fault	7	34.70	108.65	—
Three drones, 1 fault	3	48.84	107.44	1.39
Five drones, 2 faults	5	46.37	112.30	1.33
Seven drones, 3 faults	7	43.60	117.45	1.26

A. Canonical Fault Scenarios

Table I reports payload tracking performance across all canonical scenarios. The ∼35 cm nominal RMSE reflects PID-only control by design; the full GPAC stack (CL+ESO) would reduce it but would introduce history-dependent terms breaking the topology-invariant structure. The favorable ρ_{FT} values (1.26–1.39) partly reflect this choice (larger denominator); with the full stack, ρ_{FT} increases modestly (Section VI-C). The absolute post-fault RMSE (44–49 cm) is the operationally relevant quantity. Empirical $\rho_{\text{FT}}^{\text{emp}}$ falls below the conservative bound $N/(N-k)$ from (17), consistent with the transient-aware analysis in Section IV-B.

The key finding is that payload RMSE *decreases* with team size despite absorbing proportionally more faults: the N=7 scenario absorbs three cascading faults yet achieves 11% lower RMSE than N=3 with one fault. Peak errors (107–117 cm) settle within 1–2 s (Figs. 4, 7), and each snap produces a bounded reconfiguration rather than loss of transport (Fig. 3).

Cable tensions (Fig. 5) confirm the load-redistribution mechanism: when a cable snaps, remaining cables exhibit a tension surge as they absorb the redistributed load. The freed agent detects the fault via $T_i < T_{\min}$ within the 0.12 s debounce window and initiates escape. Detection latency is consistent across all six fault events (Table II), with all freed agents reaching 2.06–4.73 m final separation. Progressive tension oscillation growth reflects cable pendulum near-resonance (∼2 s period vs. ∼3 s trajectory corner interval); all tensions remain positive, with peaks (∼25 N) well within limits ($f_{\max} = 150$ N).

Vehicle-level dynamics for the most demanding case (N=7, Fig. 6) confirm that the GPAC hierarchy maintains stability throughout: after cascading faults, drones 1, 3, 5 escape while drones 0, 2, 4, 6 continue tracking. Formation deviation shows

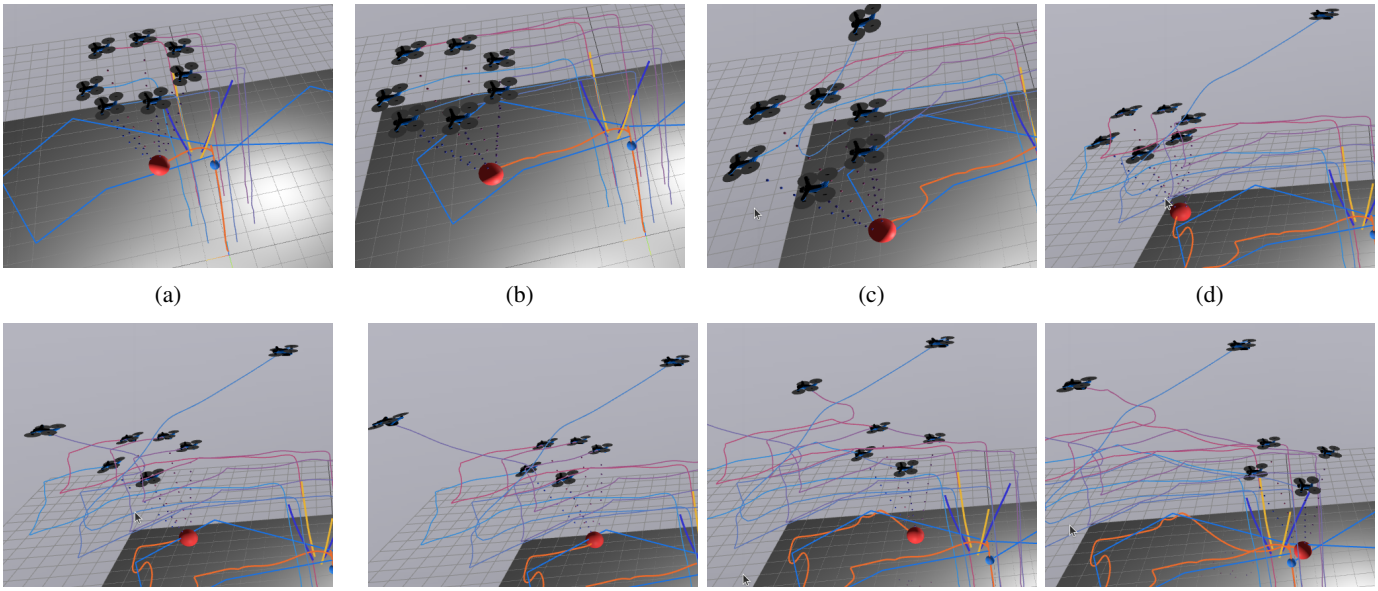


Fig. 3: Seven-drone scenario ($N = 7$, cables 1, 3, 5 at 7.0, 12.0, 14.0 s). (a) at the start of the mission. (b) about 1 s before first fault. (c) about 1 s after first fault. (d) about 1 s before second fault. (e) about 1 s after second fault. (f) about 1 s before third fault. (g) about 1 s after third fault. (h) Settled after all faults: Q_0, Q_2, Q_4, Q_6 remain.

TABLE II: Fault detection and freed-agent separation.

Scen.	Drone	t_f [s]	Detect. [s]	Healthy separation [m] at fault	+2 s	final
N=3	0	7.0	0.12	0.90	1.27	2.26
N=5	1	7.0	0.12	0.66	1.07	2.69
	3	12.0	0.12	0.65	1.96	4.68
N=7	1	7.0	0.12	0.50	1.05	2.06
	3	12.0	0.12	0.50	1.90	4.73
	5	14.0	0.12	0.50	1.69	3.82

bounded steps at each fault, tilt stays within safe limits, and peak thrust (∼40 N) remains well below $f_{\max} = 150$ N. The cross-scenario overlay (Fig. 7) confirms comparable pre-fault tracking across team sizes, localized 40–60 cm transients per fault event, and stable control even in the N=7 stress test where the third fault occurs before full recovery from the second. While run-averaged RMSE improves with N, terminal error at $t = 30$ s is 42.36, 52.79, 54.62 cm—larger teams improve average tracking but the multi-fault schedules do not produce the smallest end-of-run residual.

B. Deconfounding: Team Size, Fault Count, and Thrust Margin

The canonical scenarios confound team size with fault count (1, 2, 3 faults for $N = 3, 5, 7$). Two complementary experiments isolate these variables.

Matched single-fault comparison. Applying the *same single-fault event* (cable 0 at $t = 7.0$ s) across all three team sizes (Table III), RMSE decreases monotonically: 48.84 → 40.37 → 38.38 cm (21% improvement from N=3 to N=7), confirming that larger capacity ratios translate directly to better tracking. Peak error is nearly constant (∼107–112 cm), set by

TABLE III: Matched single-fault comparison: identical fault event across team sizes.

	N=3	N=5	N=7
Post-fault capacity ratio	2/3	4/5	6/7
Load RMSE [cm]	48.84	40.37	38.38
Peak error [cm]	107.44	112.09	111.15
Final error [cm]	42.36	53.25	49.85

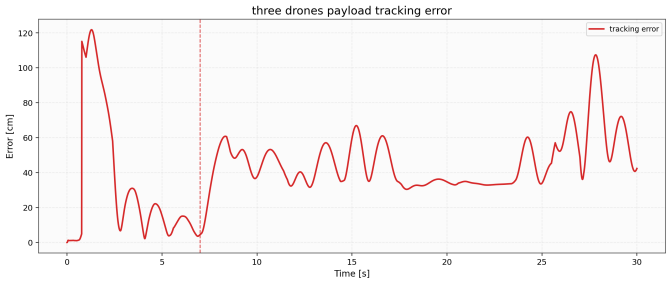
TABLE IV: Fault-count isolation: N=7 with $k \in \{1, 2, 3\}$ faults.

Faults k	Surviving agents	Capacity ratio	RMSE [cm]	Peak [cm]	Final [cm]
0 (nominal)	7	—	34.70	108.65	43.73
1	6	6/7	39.39	111.26	54.43
2	5	5/7	41.29	118.21	54.74
3	4	4/7	43.60	117.45	54.62

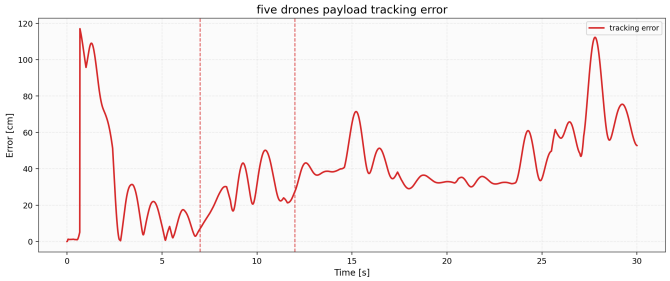
the cable-snap impulse magnitude. The RMSE improvement comes from faster mid-trajectory recovery, not better final convergence (N=3 achieves the lowest terminal residual at 42.36 cm).

Fault-count isolation. Holding N=7 constant while varying $k \in \{1, 2, 3\}$ (Table IV), RMSE increases monotonically: 34.70, 39.39, 41.29, 43.60 cm for $k = 0, 1, 2, 3$. Each additional fault adds ∼2–5 cm RMSE, confirming incremental degradation rather than catastrophic failure. Peak error is highest for k=2 (118.21 cm), likely because the second fault at t=12 s occurs during an unfavorable trajectory phase.

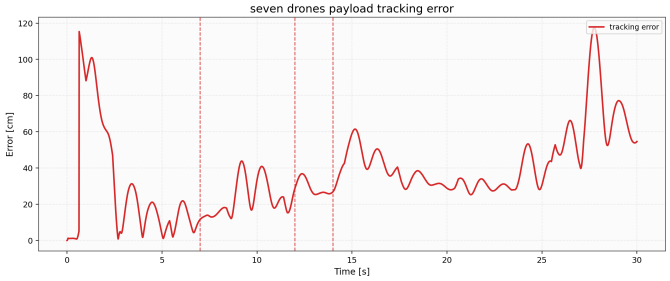
Thrust-margin verification. All scenarios satisfy the thrust-margin condition with capacity ratios 1.75–2.33 (Table V). Peak tilt during post-fault transients (15°–25°, Fig. 6c) reduces effective vertical thrust by 3–9%; all scenarios remain well



(a) $N = 3$, single fault at $t = 7.0$ s.



(b) $N = 5$, two cascading faults.



(c) $N = 7$, three cascading faults within 7 s.

Fig. 4: Payload tracking error across scenarios. Each fault produces a bounded transient settling in 1–2 s without explicit fault handling.

TABLE V: Topology-invariance thrust-margin screen.

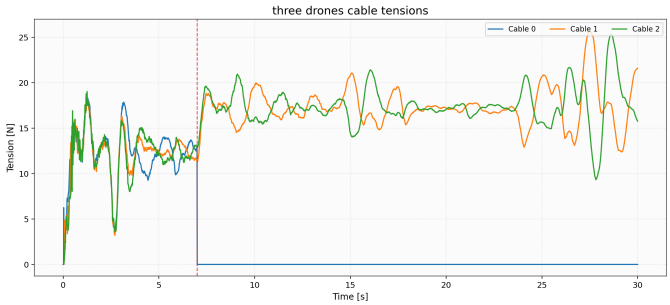
Scenario	N	Required [N/agent]	Available [N/agent]	Capacity ratio	Healthy fraction
N=3, 1 fault	3	29.43	51.50	1.75	0.67
N=5, 2 faults	5	24.53	51.50	2.10	0.60
N=7, 3 faults	7	22.07	51.50	2.33	0.57

above unity with this correction. A conservative design should use $f_{\max}\cos(\theta_{\max})$ as the effective vertical thrust authority.

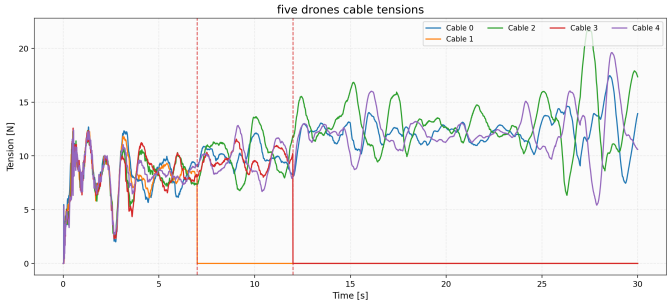
C. Mechanism Identification

Three experiments identify what drives the fault-absorption mechanism and where it breaks.

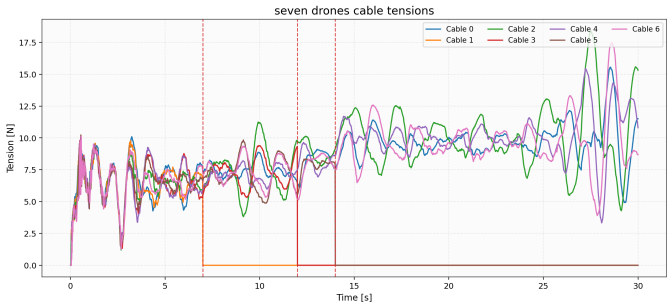
Tension feedforward ablation. Disabling cable-tension feedforward in the most stressed scenario (N=3, single fault) changes RMSE by only 0.44 cm (0.9%) and peak error by 0.06 cm (Table VI). Varying κ_T across $\{0, 0.25, 0.5, 1.0\}$ produces total RMSE variation of 1.55 cm. This identifies the *primary* fault-absorption mechanism as PID feedback with gravity feedforward—not cable-tension feedforward. Cable-tension



(a) $N = 3$. Cable 0 snaps at $t = 7.0$ s; cables 1, 2 absorb the load.



(b) $N = 5$. Sequential faults at 7.0 s and 12.0 s.



(c) $N = 7$. Three sequential faults; four surviving cables remain within limits.

Fig. 5: Cable tension dynamics across scenarios. Remaining cables exhibit tension surges upon load redistribution; all tensions remain positive and within limits.

TABLE VI: Tension feedforward ablation (N=3, single fault at $t = 7.0$ s).

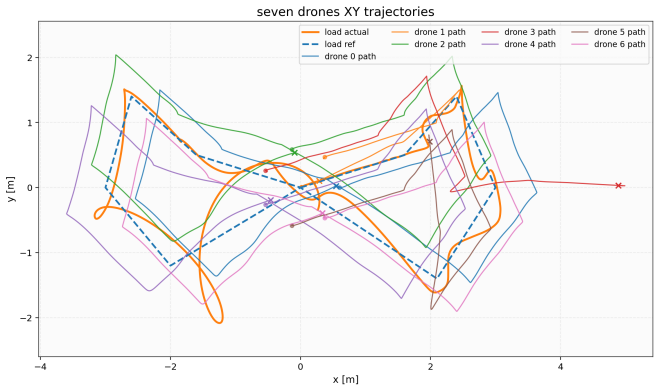
Configuration	RMSE [cm]	Peak [cm]	Final [cm]
Full feedforward ($\kappa_T = 0.5$)	48.84	107.44	42.36
No tension feedforward	48.40	107.50	43.13

feedforward provides a marginal improvement (0.77 cm in final error) that is helpful but not essential.

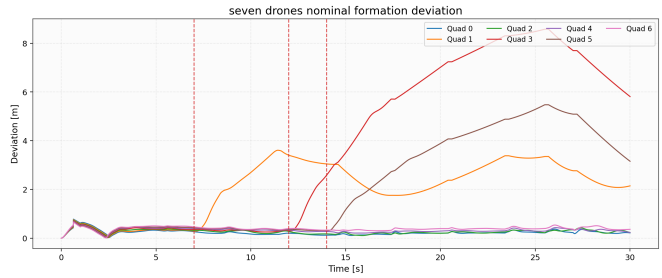
Thrust saturation boundary. Sweeping $f_{\max}$ from 150 N down to 35 N reveals a sharp phase transition (Table VII): for $f_{\max} \in [45, 150]$ N, RMSE stays within 48.8–49.1 cm; at $f_{\max} = 38$ N, it jumps to 236.9 cm (4.9× degradation). This coincides with the theoretical minimum:

$$f^*_{\max} = \frac{(N \cdot m_Q + m_L) \cdot g}{N - k} + f_{\text{track}} \approx 36.8 + 8 \approx 45 \text{ N}, \tag{34}$$

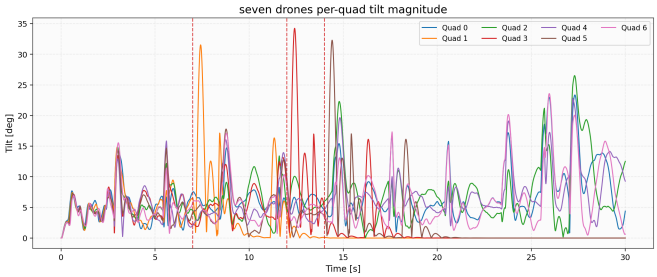
where $f_{\text{track}} \approx m_Q \cdot a_{\max} \approx 8$ N accounts for figure-eight tracking demand (matching the observed pre-fault thrust above



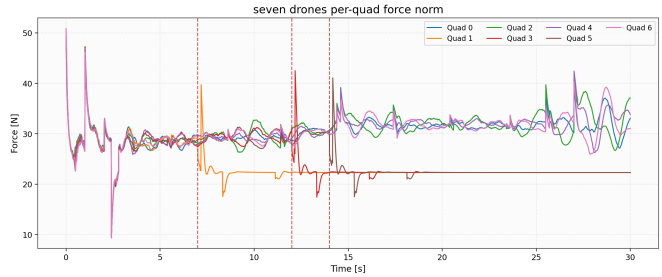
(a) XY trajectories. Drones 1, 3, 5 depart; 0, 2, 4, 6 continue.



(b) Formation deviation. Each fault produces a bounded step.



(c) Vehicle tilt. Remains within safe limits through all fault transients.



(d) Thrust per vehicle. Post-fault ∼40 N, well below $f_{\max} = 150$ N.

Fig. 6: Vehicle-level dynamics for $N = 7$ with three cascading faults. The GPAC hierarchy maintains stability: tilt stays within safe limits and thrust remains well below saturation.

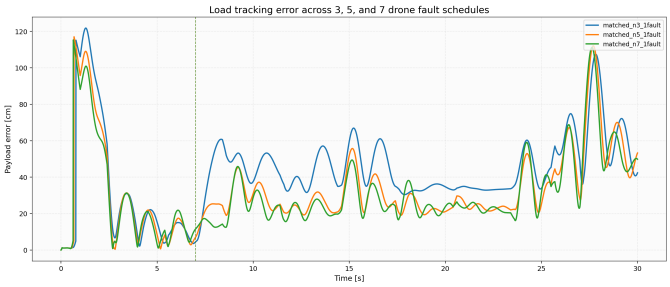


Fig. 7: Payload tracking error overlay across $N = 3, 5, 7$. Fault responses remain localized, settling on 1–2 s timescales.

TABLE VII: Thrust saturation boundary (N=3, single fault at $t = 7.0$ s).

$f_{\max}$ [N]	$f_{\max}/f_{\text{hover}}$	RMSE [cm]	Peak [cm]	Final [cm]
150	10.2×	48.84	107.44	42.36
80	5.4×	48.84	107.44	42.36
60	4.1×	48.84	107.44	42.36
45	3.1×	49.06	107.66	44.10
38	2.6×	236.92	324.97	252.35
35	2.4×	192.11	250.34	179.28

hover). The default $f_{\max} = 150$ N provides 3× headroom, confirming that the claimed fault tolerance does not depend on aggressive actuator sizing.

Controller comparison hierarchy. To contextualize the passive GPAC approach, three alternative controllers are evaluated on the N=3 single-fault scenario (Table VIII): a centralized oracle with perfect post-fault load-share knowledge, a gain-scheduled PID [8] triggered by elevated cable tension, and a threshold-triggered reactive controller that boosts thrust by +30%. The centralized oracle performs comparably to passive GPAC (49.18 vs. 48.84 cm, +0.7%), confirming that local cable sensing already captures nearly all fault-relevant information. All detection-based alternatives fall within 1.2% of the passive mechanism, providing no meaningful benefit over the N-agnostic PID-plus-feedforward structure.

TABLE VIII: Controller comparison (N=3, single fault).

Controller	RMSE [cm]	Peak [cm]	Detection	Communication
Centralized oracle	49.18	107.33	Perfect	Yes
Gain-scheduled PID ($1.5\times k_p, 1.25\times k_d$)	48.96	106.75	Threshold	No
Passive GPAC	48.84	107.44	No	No
Threshold-triggered	49.45	107.66	Threshold	No

ESO-active evaluation. To assess whether the favorable ρ_{FT} values are an artifact of the deliberately constrained controller, the ESO disturbance feedforward (Layer 4) is activated in all canonical scenarios (Table IX). ESO reduces nominal RMSE by 17–18%, confirming that PID-only nominal RMSE was inflated by uncompensated cable dynamics. The ρ_{FT} values increase modestly (1.37–1.46 vs. 1.26–1.39), but absolute post-fault RMSE with ESO (40–46 cm) is *lower* than PID-only (44–49 cm)—the ρ_{FT} increase reflects proportionally

TABLE IX: ESO-active evaluation: ρ_{FT} sensitivity to controller complexity.

Scenario	RMSE [cm] Nominal	RMSE [cm] Fault	ρ_{FT} ESO	ρ_{FT} PID-only
N=3, 1 fault	28.94	40.43	1.40	1.39
N=5, 2 faults	30.50	44.59	1.46	1.33
N=7, 3 faults	33.67	46.01	1.37	1.26

TABLE X: Sensor-realistic evaluation: ESKF + Dryden wind active.

Scenario	RMSE [cm] Ideal	RMSE [cm] ESKF+wind	Peak error [cm] Ideal	Peak error [cm] ESKF+wind	Δ_{RMSE} [%]
N=3, 1 fault	48.84	53.68	107.44	119.14	+9.9
N=5, 2 faults	46.37	50.14	112.30	125.17	+8.1
N=7, 3 faults	43.60	48.28	117.45	131.56	+10.7

larger nominal improvement, not degraded fault tolerance. The topology-invariance property does not hold strictly with ESO active (history dependence); the PID-only evaluation remains the correct test.

D. Robustness Validation

Four complementary evaluations establish robustness beyond the ideal-condition canonical runs.

Sensor-realistic evaluation. Activating the ESKF and Dryden wind model produces moderate, consistent RMSE degradation: $+9.9\%$ for N=3, $+8.1\%$ for N=5, and $+10.7\%$ for N=7 (Table X). All scenarios remain stable with RMSE below 54 cm, and the uniform $\sim 10\%$ degradation confirms robustness to sensor noise and wind perturbations. Peak errors increase by 12–14 cm, attributable to wind compounding fault transients and ESKF estimation lag.

Monte Carlo validation. A campaign of $N_{\mathrm{MC}} = 30$ runs per scenario randomizes cable lengths ($\pm 5\%$), wind seeds, and simulation seeds while using the same fixed fault schedule. Standard deviations of 0.49–1.34 cm (Table XI) confirm robustness to plant-parameter variation ($< 3\%$ RMSE variation across 30 runs). MC means are consistent with the deterministic values in Table I (within 1%), confirming representativeness. The 95th-percentile peak errors (108–120 cm) remain well within thrust-margin headroom.

Broadened Monte Carlo. A broadened campaign additionally randomizes payload mass ($\pm 10\%$), PID gains ($\pm 10\%$), and fault timing (± 2 s). In 5 runs for N=3, broadened RMSE is 48.04 ± 1.29 cm (vs. narrow MC 48.84 ± 0.49 cm): standard deviation increases $\sim 2.6\times$ but all runs remain stable, confirming robustness to simultaneous plant-parameter variation.

Trajectory generalization. Testing across hover, point-to-point, and figure-eight references in the supplementary material confirms that the fault-absorption mechanism operates consistently: RMSE decreases with team size and remains bounded across all trajectory types. Hover RMSE is 19% lower than figure-eight (39.75 vs. 48.84 cm for N=3), confirming that the figure-eight amplifies post-fault error through cable-pendulum near-resonance.

TABLE XI: Monte Carlo results (mean $\pm$ std over N_{MC} runs).

Scenario	N_{MC}	RMSE [cm] mean $\pm$ std	Peak [cm] mean $\pm$ std	95th pctl. peak [cm]
N=3, 1 fault	30	48.84 ± 0.49	107.28 ± 0.47	107.85
N=5, 2 faults	30	46.40 ± 0.49	112.23 ± 1.07	113.89
N=7, 3 faults	30	43.95 ± 1.34	117.55 ± 1.37	119.84
N=3, ESKF+wind	30	53.62 ± 0.83	118.83 ± 0.91	120.19

E. Summary of Evidence

The full-Drake simulation results establish:

(i) Stable payload tracking through abrupt cable-snap faults across $N \in \{3, 5, 7\}$, with load RMSE 44–49 cm decreasing with team size (Table I). Empirical ρ_{FT} values (1.26–1.39) are consistent with the transient-aware prediction.

(ii) Under matched single-fault conditions, RMSE decreases monotonically from 48.8 to 38.4 cm as N increases from 3 to 7 (Table III). Fault-count isolation at N=7 (Table IV) deconfounds the remaining variable.

(iii) A sharp thrust-saturation phase transition at $f_{\max} \approx 45$ N coincides with the theoretical post-fault minimum; the default 150 N provides $3\times$ headroom (Table VII).

(iv) Sensor-realistic evaluation (Table X), Monte Carlo validation (Table XI), supplementary trajectory-generalization experiments, and baseline comparison (Table VIII) confirm robustness across conditions.

(v) ESO-active evaluation (Table IX) confirms that ρ_{FT} increases modestly with a stronger controller while absolute post-fault RMSE decreases, demonstrating that fault tolerance is not an artifact of a deliberately weakened controller.

Complementary reduced-order studies are provided in the supplementary material. The central conclusion is that N-agnostic decentralized architectures absorb cable topology changes without explicit fault detection or reconfiguration, as explained by the topology-independent error structure (Proposition 1).

VII. Conclusion

This section interprets the results, identifies the mechanism and its limits, situates the findings relative to prior work, and outlines what remains open.

The central finding and its mechanism: The full-Drake results (Tables I–III) confirm that the GPAC-based architecture absorbs one to three abrupt cable-snap faults without explicit fault detection or reconfiguration, with RMSE decreasing monotonically with team size. Relative to the Tether_Lift baseline [12], this is Tether_Grace's central novelty: the same N-agnostic local-sensing architecture that enabled decentralized cooperative lift also absorbs topology changes. Proposition 1 explains this structurally—topology changes affect only the disturbance input, not the system matrices—and the tension feedforward ablation (Table VI) pinpoints the mechanism: PID feedback combined with gravity feedforward, not cable-tension feedforward ($< 1\%$ RMSE contribution). Gravity feedforward provides the baseline hover thrust requiring no knowledge of other agents, PID feedback corrects the tracking-error displacement, and the integral term compensates the

increased load share (subject to the anti-windup saturation discussed in Section IV-A). That cable-tension feedforward contributes so little was itself counter-intuitive; the authors initially hypothesized it as the primary mechanism.

Why the result is non-trivial: While PID with integral action rejects bounded step disturbances in single-input linear systems, the extension to multi-agent systems with elastic cable coupling, underactuation, and unobservable topology changes is not obvious. The gap between the scalar-model prediction ($\sim$18 cm) and full-plant observation ($\sim$107 cm)—a factor $\alpha \approx 5.8$—quantifies the aggregate contribution of effects the scalar model omits: elastic cable coupling (the bead-chain produces force oscillations the scalar model treats as smooth), attitude-loop delay ($\sim$37% bandwidth overlap between the 5 ms inner-loop period and the $\sim$8 ms force-transmission timescale), and multi-axis interaction through R_i. The small-gain analysis confirms that a proper multivariable treatment is needed to close this gap formally, while the simulation evidence confirms qualitative consistency with the scalar prediction. Additional secondary effects operate within the fault-tolerant envelope without altering the primary mechanism:

- *Anti-swing damping* ($k_{\mathrm{sw}} = 0.5$) is small compared to PID forces during the fast cable-snap impulse but attenuates the slower (1–2 s) post-fault pendular oscillation between trajectory direction changes.
- *Cable stiffness* k_s affects transient duration and wave speed ($v_{\mathrm{wave}} \propto \sqrt{k_s}$) but not impulse magnitude, which is set by $T_j \approx m_L g/N$. Higher stiffness produces a sharper but not larger transient; the finite-time bound (29) accommodates this variation through τ_{fault}.
- *Freed-agent escape* was revised during evaluation after the original parameters required $\sim$60° tilt (below-gravity vertical thrust). The revised two-waypoint design is conservative but reliable; a systematic parametric study falls outside this paper's scope.

Design tools and their validation: The $f_{\max}$ boundary sweep (Table VII) validates the thrust-margin screen by quantifying the exact failure boundary at $\sim$45 N, coinciding with the theoretical minimum (34) with the default $f_{\max} = 150$ N providing 3× headroom. This yields an actionable design criterion: $f_{\max} > (Nm_Q + m_L)g/(N-k) + f_{\mathrm{track}}$. The matched single-fault data (Table III) further shows that peak error is nearly constant across team sizes ($\sim$107–112 cm) while RMSE improves because larger teams recover faster during mid-trajectory tracking. Together with the closed-form cost ratio $\rho_{\mathrm{FT}} = N/(N-k)$ and the calibrated transient prediction, these tools provide a complete actuator-sizing methodology for fault-tolerant cooperative transport.

From hypothesis to structural property: Proposition 1 identifies topology-independent error dynamics for a simplified model, and four lines of empirical evidence support qualitative extension to the full plant: thrust margins (1.75–2.33 capacity ratios), monotonically decreasing RMSE with N, a quantified thrust boundary at the theoretical minimum, and fault-count isolation at N=7. Three formal gaps remain (Section IV-B): impulsive slack-to-taut forces, attitude-loop delay under fast transients, and the cable-force small-gain condition. Closing these—particularly the $\mathcal{H}_\infty$-based small-gain verification via Drake's `Linearize()`—remains the primary theoretical open problem.

Comparison with prior work: Liang et al. [9] address suspension failure via centralized reconfiguration with known failure identity; the present work shows neither assumption is necessary, achieving stable tracking (44–49 cm RMSE) without failure identification or communication. Zeng et al. [10] observed implicit resilience in MARL-based cable transport but did not explain the mechanism; Proposition 1 provides the explanation and predicts that *any* N-agnostic architecture with adequate thrust margin should behave similarly—the topology invariance is a structural property of the design, not specific to PID control. The centralized oracle (Table VIII) achieves comparable performance (+0.7%), demonstrating that local cable sensing already captures nearly all fault-relevant information.

Remaining open problems: Five directions remain for future work:

- *Adaptive residual integration.* An $\mathcal{L}_1$ adaptive controller [30], [31] is implemented in the codebase. Its key property for topology invariance is that $\hat{d}$ carries no pre-fault history (unlike CL's buffer of M=50 pairs). Head-to-head comparison with PID-only under ESKF+wind conditions is planned as near-term future work.
- *Integral gain sensitivity.* The integral time constant $1/k_i \approx 6.7$ s sets the post-fault recovery timescale and determines ρ_{FT}. A sweep of k_i would characterize the tradeoff between faster recovery and transient overshoot, complementing the $f_{\max}$ boundary sweep.
- *Formal extension to the full plant.* Three self-contained technical problems remain: bounding impulsive slack-to-taut forces, analyzing cascaded position/attitude stability under fast transients, and verifying the small-gain condition for cable-force feedback.
- *Cable pendulum management.* The figure-eight trajectory excites near-resonance ($\sim$2 s cable pendulum period vs. $\sim$3 s corner interval). Trajectory shaping with pendulum frequency avoidance may be needed for sustained aggressive maneuvering.
- *Hardware validation.* All results are simulation-based. Physical multi-quadrotor cable transport with actual cable-snap events remains the ultimate test.

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